CH 5 — Differentiability

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Recall $f(x,y) = \begin{cases} \frac{2xy}{x^2+y^2} & (x,y)\neq 0 \\ 0 & (x,y)=(0,0) \end{cases}$

We have previously shown that f is not continuous at $(0,0)$

But $f_x(0,0) = f_{y(0,0)} = 0$. Thus partial derivatives exists

Notice that here differentiability does not imply continuity

Recall in Calculus I where cusps, jump discontinuity, vertical asymptotes are not differentiable

When $f(x)$ differentiable, the linear approximation is defined and is a good local approximation on the graph

> 📘 **Definition — Remainder Approximation**
>
> If $g'(a)$ exists, then
>
> $$\lim_{x\to a} \frac{|g(x) - L_a(x)|}{|x-a|} = \lim_{x\to a} \frac{|R_{1,a}(x)|}{|x-a|}$$

Proof

$R_{1,a}(x) = g(x) - L_a(x) = g(x) - [g(a) + g'(a)(x-a)]$

$\frac{|R_{1,a}(x)|}{|x-a|} = \frac{|\ g(x)-[g(a)+g'(a)(x-a)]|}{|x-a|} = |\frac{g(x)-g(a)}{x-a} - g'(a)|, \quad x \neq a$

$\lim_{x\to a} \frac{|R_{1,a}(x)|}{|x-a|} = \lim_{x\to a} |\frac{g(x)-g(a)}{x-a} - g'(a)| = |g'(a) - g'(a)| = 0$

□

> 📘 **Definition — Differentiability**
>
> $f(x)$ is differentiable at $x = a$ if
>
> $$\lim_{x\to a} \frac{|R_{1,a}(x)|}{|x-a|} = 0$$
>
> $f(x,y)$ is differentiable at $(x,y) = (a,b)$ if
>
> $$\lim_{(x,y)\to(a,b)} \frac{|R_{1,a}(x,y)|}{\|(x,y)-(a,b)\|} = 0$$

Note: for the linearization to exist at (a,b), both partial derivatives must exists at (a,b) but this is not enough to guarantee differentiability. The existence of the partial derivatives is a necessary condition, but not sufficient condition for differentiability.

Example:

Show that $g(x) = x^2$ satisfy the above definition on differentiability at $x = 1$.

$R_{1,a} = g(x) - [g(a) + g'(a)(x-a)] = x^2 - 1 - 2(x-1) = x^2 - 2x + 1$

$\lim_{x\to 1} \frac{|x^2-2x+1|}{|x-1|} = \lim_{x\to a} |\frac{x^2-2x+1}{x-1}| = \lim_{x\to a} |x-1| = 0$

Definition — Derivatives at a Point

If $f(x,y)$ satisfy

$$\lim_{(x,y)\to(a,b)} \frac{|f(x,y) - f(a,b) - c(x-a) - d(y-b)|}{\|(x,y)-(a,b)\|} = 0$$

then $c = f_{x(a,b)}, d = f_{y(a,b)}$

Proof

Let $\vec{z} = \{(a,b) + c(x-a) + d(y-b)\}$ be any plane $(a.b, f(a,b))$

We assume $\lim_{(x,y)\to(a,b)} \frac{|f(x,y)-f(a,b)-c(x-a)-d(y-b)|}{\|(x,y)-(a,b)\|} = 0$

This must be true on all paths approaching (a,b)

If $y = b, \lim_{x\to a} \frac{|f(x,b)-f(a,b)-c(x-a)|}{|x-a|} = 0 \implies \lim_{x\to a} |\frac{f(x,b)-f(a,b)|}{x-a} - c| = 0 \iff c = f_x(a,b)$

If $x = a, \lim_{y\to b} \frac{|f(a,y)-f(a,b)-d(y-b)|}{|y-b|} = 0 \implies \lim_{y-b} |\frac{f(a,y)-f(a,b)|}{y-b} - d| = 0 \iff d = f_y(a,b)$

$\vec{z} = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$

□

Definition — Tangent Plane

Consider a function $f(x,y)$ which is differentiable at (a,b). The **tangent plane** to the surface $z = f(x,y)$ at (a,b) is

$$z = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$$

Examples:

1. Determine whether $f(x,y) = x^2 + y^2$ is differentiable at $(0,0)$

   $f_{x(0,0)} = \lim_{h\to 0} \frac{f(0,h)-f(0,0)}{h} = \frac{0}{h} = 0; f_{y(0,0)} = \lim_{h\to 0} \frac{f(h,0)-f(0,0)}{h} = 0$

   $L_{0,0} = f(0,0) + f_x(0,0)(x) + f_y(0,0)(y) = 0 + 0 + 0 = 0$

   $\lim_{(x,y)\to(0,0)} \frac{|R_{1,(0,0)}(x,y)|}{\|(x,y)-(0,0)\|} = \lim_{(x,y)\to(0,0)} \frac{|x^2+y^2|}{\sqrt{x^2+y^2}} = \lim_{(x,y)\to(0,0)} \sqrt{x^2+y^2} = 0$

   f is differentiable at the origin

2. $f(x,y) = \begin{cases} \frac{2xy}{x^2+y^2} & \text{if } (x,y)\neq(0,0) \\ 0 & \text{if } (x,y)=(0,0) \end{cases}$ is not differentiable at $(0,0)$

   $f_x(0,0) = 0 = f_y(0,0); L_{0,0}(x,y) = f(0,0) + f_x(0,0)(x) + f_y(0,0)(0) = 0+0+0 = 0$

   $\lim_{(x,y)\to(0,0)} \frac{|R_{1,(0,0)}|}{\|(x,y)-(a,b)\|} = \lim_{(x,y)\to(0,0)} \frac{\frac{|2xy|}{x^2+y^2}}{\sqrt{x^2+y^2}} = \lim_{(x,y)\to(0,0)} \frac{2|x\|y|}{(x^2+y^2)^{\frac{3}{2}}}$

   Try approach $(0,0)$ along path $y = \sqrt{x} \implies \lim_{x\to 0} \frac{2}{(x+1)^{\frac{3}{2}}} = 2 \neq 0.$

   $\therefore f$ is not differentiable at $(0,0)$

3. $f(x,y) = |xy|$ is differentiability at $(0,0)$ but not at $(1,0)$

   $f_x(0,0) = \lim_{h\to 0} \frac{f(h,0)-f(0,0)}{h} = \lim_{h\to 0} \frac{0}{h} = 0$; $f_y(0,0) = \lim_{h\to 0} \frac{f(0,h)-f(0,0)}{h} = \lim_{h\to 0} \frac{0}{h} = 0$

   $L_{0,0}(x,y) = 0$

   $\lim_{(x,y)\to(0,0)} \frac{|R_{1,(0,0)}|}{\|(x,y)-(0,0)\|} = \lim_{(x,y)\to(0,0)} \frac{|xy|}{\sqrt{x^2+y^2}}$

   $|\frac{xy}{\sqrt{x^2+y^2}}| \leq \frac{\sqrt{x^2+y^2}|y|}{\sqrt{x^2+y^2}} = |y|; \lim_{y\to 0}|y| = 0$

   By Squeeze Theorem, $\lim_{(x,y)\to(a,b)} \frac{|xy|}{\sqrt{x^2+y^2}} = 0$

   $f_x(1,0) = \lim_{h\to 0} \frac{f(x+h,0)-f(x,0)}{h} = \lim_{h\to 0} \frac{0}{h} = 0$

   $f_y(1,0) = \lim_{h\to 0} \frac{f(1,y+h)-f(1,y)}{h} = \lim_{h\to 0} \frac{|h|}{h} = \text{DNE}$

   $f_y(1,0)$ does not exist, thus the linearization of $f(x)$ does not exist. Therefore, $f(x,y)$ is not differentiable at $f(1,0)$

> 📘 **Definition — Differentiability and Continuity**
>
> If $f(x,y)$ is differentiable at (a,b), then $f(x,y)$ is continuous at (a,b)

Proof

Using the fact that the $R_{1,(a,b)}(x,y) = f(x,y) - L_{a,b}(x,y)$, we can write

$$\begin{aligned} f(x,y) &= L_{a,b}(x,y) + R_{1,(a,b)} \\ &= f(a,b) = \nabla f(a,b)\cdot(\vec{x}-\vec{a}) + R_{1,(a,b)}(x,y) \end{aligned}$$

$$\begin{aligned} \lim_{(x,y)\to(a,b)} f(x,y) &= \lim_{(x,y)\to(a,b)} f(a,b) + \lim_{(x,y)\to(a,b)} \nabla f(a,b)(x-a,y-b) + \lim_{(x,y)\to(a,b)} R_{1,(a,b)}(x,y) \\ &= f(a,b) + 0 + \lim_{(x,y)\to(a,b)} R_{1,(a,b)} \end{aligned}$$

For $(x,y) \neq (a,b)$:

$$R_{1,(a,b)} = \frac{R_{1,(a,b)}\ \|(x,y)-(a,b)\|}{\|(x,y)-(a,b)\|}$$

since f is differentiable at $(0,b)$,

$$\lim_{(x,y)\to(a,b)} R_{1,(a,b)} = \lim_{(x,y)\to(a,b)} \frac{R_{1,(a,b)}\ \|(x,y)-(a,b)\|}{\|(x,y)-(a,b)\|} = 0$$

$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b) + 0 + 0$

$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$

□

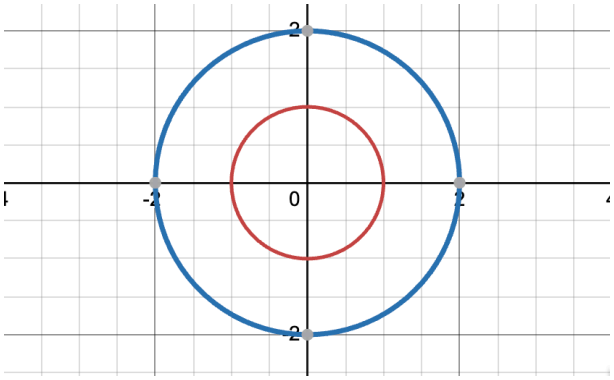
Examples

1. $f(x,y) = \sqrt{x^2+y^2}$

   Level curves:

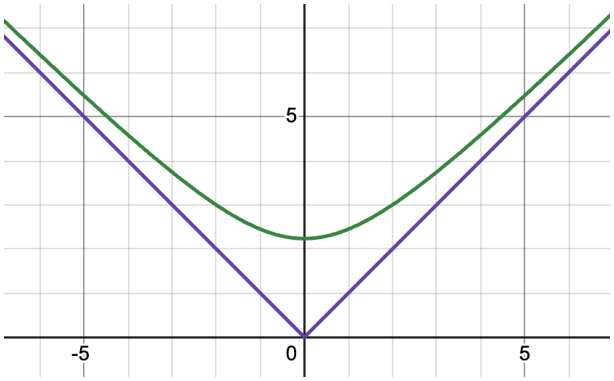
   $\sqrt{x^2+y^2} = k, \quad k \geq 0$

   $x^2 + y^2 = k^2$

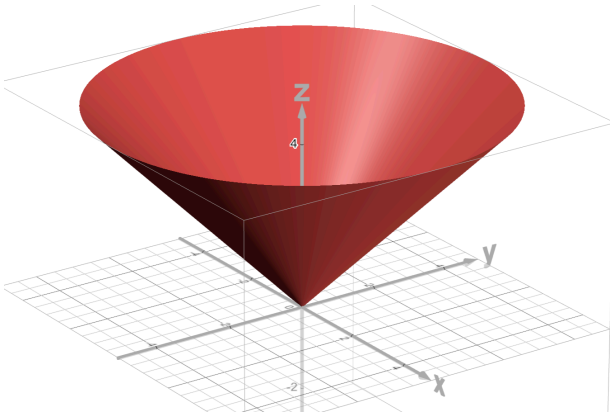


   Cross Secions

   $z = \sqrt{c^2+y^2}, c \in \mathbb{R}, z = \sqrt{x^2+d^2}, d \in \mathbb{R}$



   Graph in 3D



2. Show the above function is not differentiable at $(0,0)$

   $f_x = \lim_{h\to 0} \frac{f(h,0)-f(0,0)}{h} = \lim_{h\to 0} \frac{\sqrt{h^2}}{h} = \lim_{h\to 0} \frac{|h|}{h} = \text{DNE}$

   $f_x(0,0)$ does not exist, thus the linearization does not exist at $(0,0)$. So $f(x,y)$ is not differentiable at $(0,0)$

Theorem — Mean Value Theorem

If $f(t)$ is contnuous on the closed interval $[t_1,t_2]$, f is differentiable on the open interval (t_1,t_2), then there exists $t_0 \in (t_1,t_2)$ such that

$$f(t_2)-f(t_1) = f'(t_0)(t_2-t_1)$$

Theorem — Continuity Theorem

If the partial derivatives $f_x(x,y)$ and $f_y(x,y)$ are continuous at (a,b), then $f(x,y)$ is differentiable at (a,b)

Proof

$R_{1,(a,b)} = f(x,y) - f(a,b) - f_x(a,b)(x-a) - f_y(a,b)(y-b)$

f_x, f_y are continuous at (a,b), so f_x, f_y are defined in some neighbourhood of (a,b), $N(a,b)$ For $(x,y) \in N(a,b)$

$$\begin{aligned}
f(x,y)-f(a,b) &= [f(x,y)-f(a,y)] + [f(a,y)-f(a,b)] \\
[f(x,y)-f(a,y)] &= f_x(\overline{x},y)(x-a) \\
&= f_x(\overline{x},y)(x-a) + f_x(a,b)(x-a) - f_x(a,b)(x-a) \qquad \overline{x}\in(x,a) \\
&= f_x(a,b)(x-a) + A(x-a) \\
\implies A &= f_x(\overline{x},y) - f_x(a,b) \\
[f(x,y)-f(a,y)] &= f_y(a,\overline{y})(y-b) \\
&= f_x(a,\overline{y})(y-b) + f_y(a,b)(y-b) - f_y(a,b)(y-b) \qquad \overline{y}\in(y,b) \\
&= f_y(a,b)(y-b) + B(y-b) \\
\implies B &= f_y(a,\overline{y}) - f_y(a,b)
\end{aligned}$$

$$\begin{aligned}
R_{1,(a,b)}(x,y) &= f_x(a,b)(x-a) + A(x-a) + f_y(a,b)(y-b) + B(y-b) - f_x(a,b)(x-a) - f_y(a,b)(y-b) \\
&= A(x-a) + B(y-b)
\end{aligned}$$

$$\begin{aligned}
\left|\frac{|R_{1,(a,b)}(x,y)|}{\|(x,y)-(a,b)\|} - 0\right| &= \left|\frac{|R_{1,(a,b)}(x,y)|}{\|(x,y)-(a,b)\|}\right| = \frac{|A(x-a)+B(y-b)|}{\sqrt{(x-a)^2+(y-b)^2}} = \frac{|A\|x-a|}{\left(\sqrt{(x-a)^2+(y-b)^2}\right)} + \frac{|B\|y-b|}{\left(\sqrt{(x-a)^2+(y-b)^2}\right)} \\
&= \frac{|A|\sqrt{(x-a)^2}}{\left(\sqrt{(x-a)^2+(y-b)^2}\right)} + \frac{|B|\sqrt{(y-b)^2}}{\left(\sqrt{(x-a)^2+(y-b)^2}\right)} \le \frac{|A|s\left(\sqrt{(x-a)^2+(y-b)^2}\right)}{\left(\sqrt{(x-a)^2+(y-b)^2}\right)} + \frac{|B|\left(\sqrt{(x-a)^2+(y-b)^2}\right)}{\left(\sqrt{(x-a)^2+(y-b)^2}\right)} \\
&= |A| + |B|
\end{aligned}$$

Since f_x, f_y is continuous at (a,b), $\lim_{(x,y)\to(a,b)} f_x(\overline{x},y) = f_x(a,b)$,

$\lim_{(x,y)\to(a,b)} f_y(a,\overline{y}) = f_y(a,b)$

$\implies \lim_{(x,y)\to(a,b)} A = 0, \lim_{(x,y)\to(a,b)} B = 0 \implies \lim_{(x,y)\to(a,b)} |A| + |B| = 0$

By Squeeze Theorem, $\lim_{(x,y)\to(a,b)} \frac{|R_{1,(a,b)}(x,y)|}{\|(x,y)-(a,b)\|} = 0$ so f is differentiable at (a,b)

□

Example:

$f(x,y) = \sqrt{x^2+y^2}$ is differentiable $\forall (x,y) \neq (0,0)$

For $(x,y) \neq (0,0)$

$f_x = \frac{x}{\sqrt{x^2+y^2}}, f_y = \frac{y}{\sqrt{x^2+y^2}}$

By continuity theorem, f_x, f_y are continuous $(x,y) \neq (0,0)$, therefore differentiable $(x,y) \neq (0,0)$

Notation: if the k-th order partial derivatives of $f(\vec{x})$ are continuous, $f \in C^k$

If $f(x,y) \in C^k \implies f_x, f_y$ are continuous therefore f is differentiable

$f(x,y) = \begin{cases} (x^2+y^2)\sin\left(\frac{1}{\sqrt{x^2+y^2}}\right) \text{ if } (x,y)\neq(0,0) \\ 0 \text{ if } (x,y)=(0,0) \end{cases}$ Notice f is differentiable at $(0,0)$ and has a horizontal tangent plane. But f_x, f_y are discontiuous due to rapid oscillation